

# Caio Valerio Goulart Correia

Backend Engineer (TypeScript/Node.js) · APIs & Developer Tools · AI Specialist

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Backend engineer with 9+ years of technical experience building and maintaining APIs, services, and developer-facing tools — primarily TypeScript and Node.js, with deep Python alongside. Shipped and operates a production HR/operations platform (200+ employees) whose Node.js/TypeScript backend exposes 600+ API routes secured by JWT, two-factor authentication, and role-based authorization (PostgreSQL RLS + ACL). Defines cross-product API contracts (HMAC-authenticated sync), hardens third-party integrations with retries and fallback on rate-limit/503 responses, and maintains open-source developer tools on GitHub. Also deep in AI: LLM training/evaluation (SFT, DPO/RLHF), local serving (vLLM, SGLang), and RAG. Fluent English (C1), remote-first with international teams.

## SKILLS

TypeScript · Node.js · Express · Next.js · REST APIs · API Design · GraphQL · JWT & Auth · Python · FastAPI · PostgreSQL · Supabase · PostGIS · Docker · Kubernetes · AWS · GitHub Actions · CI/CD · Linux · Git · Java · Go · Generative AI · LLMs · RAG · Model Fine-Tuning · MLOps · vLLM · MCP · Backend Development · Data Engineering · Security · Compliance

## PROFESSIONAL EXPERIENCE

### AI Specialist (Contract)

micro1 · Aug 2026 – Sep 2026

- Trained and evaluated LLMs for frontier-model programs, running Supervised Fine-Tuning (SFT) and preference optimization (DPO/RLHF) with PyTorch, Hugging Face PEFT (LoRA/QLoRA), Unsloth, and Axolotl.
- Ran adversarial testing (red teaming) to surface hallucinations, logic flaws, and bias, feeding corrections back into training datasets.
- Reviewed and benchmarked AI-generated code across Python, TypeScript, Node.js, FastAPI, React, Next.js, and JavaFX — assessing data-structure choices, execution behavior, and complex API integrations.
- Deployed local LLM serving and agent harnesses on Windows/Linux (vLLM, SGLang, llama.cpp) on dedicated NVIDIA hardware (L4 GPU, 24GB VRAM), plus MCP tool servers for retrieval and graph tools; measured latency/throughput with custom evaluation harnesses.
- Curated high-quality training and synthetic datasets (Self-Instruct, Evol-Instruct) with automated validation for dataset integrity, schema compliance, and model alignment.

### Full-Stack Developer & DevOps Engineer

ABZ Serviços · Mar 2025 – Present

- Architected and shipped EmployeeHub ([portal.groupabz.com](https://portal.groupabz.com)), an HR/operations portal used by 200+ employees — personnel records, crew management, e-Social, vacation, reimbursement, and ACL — on Next.js 15, TypeScript, and Supabase (PostgreSQL with Row-Level Security) on Vercel; reduced HR manual workload by ~40% and approval time by ~95%.
- Built and maintains the platform's Node.js/TypeScript backend — 600+ REST API routes (Next.js App Router) with JWT authentication, two-factor authentication (2FA), and role-based authorization via PostgreSQL Row-Level Security and ACL, protecting sensitive HR/PII data under LGPD.
- Delivered the ABZ product suite around the portal: PontoFlow (bilingual offshore timesheets with manager approval, field annotations, and deadline locks enforced in UI and RLS), Invoice ABZ (FastAPI/SQLAlchemy engine turning payroll and charge-rate spreadsheets into client invoices), and Ticket-Manager (support automation via Microsoft Graph).
- Defined the cross-product API contract (HMAC-authenticated import/export) between the portal and the PontoFlow timesheet product, keeping collaborator identity and period behavior consistent across both systems.
- Led the cloud migration of legacy on-premise Windows Servers (2012–2025), reaching 80% completion with zero production downtime.
- Implemented end-to-end CI/CD with GitHub Actions and Docker, administered Proxmox virtualization and AWS S3, and deployed a WireGuard VPN for 50+ remote employees — sharply reducing connectivity support tickets.

## Python Developer & Spatial Data Engineer

BRTech3D · Mar 2024 – Mar 2025

- Built high-performance Python pipelines (NumPy, SciPy, Open3D) automating geospatial data processing for dense point clouds (100M+ points), increasing processing speed by 40% and improving data compression across dozens of projects.
- Modeled and optimized PostgreSQL databases with the PostGIS extension for complex spatial queries, designing and maintaining the REST APIs that exposed them to internal processing pipelines.

## Topographical Survey Operator III (Laser Scanning)

Laser Master Engineering Services · Aug 2023 – Mar 2024

- Operated high-precision 3D laser scanning equipment (Leica) in complex industrial and archaeological environments.
- Led field campaign planning and technical documentation, ensuring millimeter-level precision and data backup redundancy for large-scale data collection.

## Topographical Survey Operator II

BRTech3D · Dec 2022 – Aug 2023

- Executed field data collection using Leica TLS equipment and introduced Quality Assurance (QA) processes for outlier cleaning and multiple scan registration.

## Technical Documentation & Support Specialist

SmartStore / Jackkk Connection · Mar 2021 – Dec 2022

- Produced 50+ bilingual technical manuals and system specifications for international clients.
- Provided advanced pre-sales support and detailed hardware/software architecture mapping for complex business proposals.

## Training Specialist & Automation Technician

Jackkk Connection Comercial · May 2017 – Mar 2021

- Coordinated and delivered technical training for over 500 professionals in PLCs, hydraulics, and industrial automation.
- Developed educational simulations and managed B2B technical sales, building direct relationships with international suppliers.

## Technical Team Lead & Specialized Support

Diesel Line Cambuí · Aug 2016 – May 2017

- Led specialized field teams in the maintenance of large-scale engines and turbines (CAT, Cummins, MAK).
- Managed English-language technical support tickets for international clients and formally documented failure analysis processes.

## Apprentice Technician

Diesel Line Cambuí · Oct 2015 – Aug 2016

- Began career providing support to commercial processes, CRM management, and strict application of compliance procedures for diesel and gas systems maintenance.

## EDUCATION

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### B.S. in Mechanical Engineering

UniFatecie

### Technical Diploma in Industrial Automation

ETP Escola Técnica Brasil

## RELEVANT PROJECTS

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### Painel ABZ

Next.js 15 · TypeScript · PostgreSQL · Supabase

Corporate operations portal for ABZ Group: HR, offshore crew management, payroll, reimbursements and internal comms on Next.js 15, React 18 and Supabase/PostgreSQL (50+ tables with RLS) behind a Node.js/TypeScript API layer of 600+ routes with JWT auth and 2FA. Ships e-Social (13 events, RSA-SHA256 XML, SOAP mTLS), OCR (Tesseract) for ASO/ID docs, digital signatures, MIO ERP sync, and an AI companion with LiveKit voice, MS Graph and autonomous KPI agents.

<https://github.com/Caiolino00/EmployeeHub>

### PontoFlow

Next.js 15 · TypeScript · Node.js · Supabase

Bilingual (pt-BR/en-GB) timesheet system for offshore crews with manager approval, field-level annotations, and monthly deadline locks enforced in UI and Postgres RLS. Node.js server runtime with Nodemailer cron reminders and multi-tenant admin; syncs with EmployeeHub through an HMAC-authenticated import/export API contract.

<https://github.com/Caiolino00/PontoFlow>

## CloudSec & FinOps Auditor

Next.js · TypeScript · RAG · Gemini

Compliance auditor that evaluates cloud architecture scenarios against CIS / SOC 2 / FinOps-style policies, returning structured risk, cost impact, citations, and remediation steps. Hybrid RAG (BM25 + TF-IDF + reciprocal rank fusion) over versioned policies; resilient LLM calls with automatic retry and model fallback on 503/rate-limit responses; Vitest tests and CI evaluation pipeline.

<https://github.com/Caiolino00/cloudsec-finops-auditor>

## SGLang Commander

Python · FastAPI · React · PySide6

Open-source desktop (PySide6) and web (React + FastAPI) console for SGLang inference servers. Starts and stops servers, streams multimodal chat (image/audio), plots live GPU/VRAM/latency metrics, deploys Hugging Face models in one click, runs P50/P95/P99 benchmarks, and supports Docker plus ZeroTier remote deploys.

<https://github.com/Caiolino00/sglang-commander>

## Visualizador 3D FARO

Three.js · React · Open3D · MCP

Matterport-style web viewer for FARO Focus scans. Renders PTS/E57 point clouds and TrueView 360 panoramas with Three.js and React Three Fiber, plus dollhouse view, floor plans, distance measurement, annotations, auto-tours and a Model Context Protocol bridge so local LLMs can drive the 3D scene.

[https://github.com/Caiolino00/Visualizador3D\\_Matterport\\_Clone](https://github.com/Caiolino00/Visualizador3D_Matterport_Clone)

## Gerador de Malha Open3D

Python · Open3D · NumPy

Desktop pipeline that turns industrial point clouds into production meshes: Poisson reconstruction, hole filling, density-based vertex cleanup, smoothing and export to PLY/OBJ/STL — built for geospatial and laser-scan workflows with 100M+ points.

[https://github.com/Caiolino00/Projeto\\_Open\\_Mesh\\_](https://github.com/Caiolino00/Projeto_Open_Mesh_)

## AD Migration Suite

C# · .NET 8 · WPF

Enterprise Active Directory migration from Windows Server 2012/2016 to 2019/2022. Agent-based architecture on a single port (8765) avoids WinRM/Kerberos setup, preserves SID history, replicates GPOs/OU's and migrates file shares with ACL translation across domains.

[https://github.com/Caiolino00/AD\\_Migrator](https://github.com/Caiolino00/AD_Migrator)

## Auditoria de Dados

Python · Pandas · Dash · Plotly

Server data-audit toolkit with parallel directory scans, file-type and growth analysis, caching and an interactive Dash/Plotly dashboard. Generates timestamped HTML and XLSX reports with client-level filters for large file-server inventories.

[https://github.com/Caiolino00/Python\\_Audit\\_Script](https://github.com/Caiolino00/Python_Audit_Script)

## Network Monitor

Python · Flask · PWA

Cross-platform network monitor (Windows, Linux, macOS, including Windows Server 2012) with live download/upload/latency, speed tests, history charts, PDF reports and PWA install for offline use on the LAN.

<https://github.com/Caiolino00/network-monitor>

## TECHNICAL SKILLS

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### AI & LLM Engineering

Generative AI, LLMs, RAG, Prompt Engineering, Model Fine-Tuning (SFT, DPO/RLHF, LoRA/QLoRA), MLOps, vLLM, SGLang, llama.cpp, MCP.

### Backend & APIs

TypeScript, Node.js (Next.js server runtime, Express, REST API routes), Python (FastAPI, SQLAlchemy), API design & cross-product API contracts, authentication & authorization (JWT, 2FA, RLS, ACL, HMAC), resilience patterns (retries, fallback, rate-limit/503 handling), GraphQL, Java (Spring Boot), Go.

### Cloud & DevOps

Linux, Git, DevOps, Docker, Kubernetes, AWS (S3), GitHub Actions (CI/CD), Vercel, Cloudflare, Proxmox, WireGuard VPN, Terraform.

### Databases

PostgreSQL, PostGIS, Supabase (Row-Level Security), SQLite (FTS5), NoSQL, ETL/data pipelines.

**Governance, Security & Compliance**

Data Governance, Security, Compliance, LGPD/PII controls.

**LANGUAGES**

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Portuguese: Native | English: Fluent (C1) | Spanish/Italian: Advanced technical reading.